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Emergence of functional information from multivariate correlations

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The information content of symbolic sequences (such as nucleic or amino acid sequences, but also neuronal firings or strings of letters) can be calculated from an ensemble of such sequences, but because information cannot be assigned to single sequences, we cannot correlate information to other observables attached to the sequence. Here we show that an information *score* obtained from multivariate (multiple-variable) correlations within sequences of a 'training' ensemble can be used to predict observables of out-of-sample sequences with an accuracy that scales with the complexity of correlations, showing that functional information emerges from a hierarchy of multi-variable correlations.

This article is part of the theme issue 'Emergent phenomena in complex physical and socio-technical systems: from cells to societies'.

1. Introduction

The term 'functional information' refers to the information that is necessary and sufficient to predict the function of a symbolic sequence within a particular environment. Typically, the sequence could be nucleic or amino acids, but it also could be a sequence of neuronal firings or even a string of letters written in a particular alphabet. The function could be the biochemical activity (for biological sequences), or the level of performance for a neuronal network that produces the firings. For words in a language, the functionality could be simply the likelihood that the word is understood by someone familiar with the language.

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We often simply use the word 'information' to indicate functional information. For example, the DNA sequence that carries the information necessary to make the protein haemoglobin is functional in the sense that (given the right cellular environment) it binds oxygen with a particular affinity. For recordings from animal brains, the functionality of the sequence of neuronal activations could be the accuracy with which the animal performs a rewarded task.

Functional information can be defined quantitatively in terms of the likelihood that a *random* sequence performs the function or task at the specified level [1,2]. Specifically, if we require a particular sequence to perform function E at least at the level θ, then functional information about E as defined by Szostak is given by

$$I(E_\theta) = -\log F(E \geq \theta), \tag{1.1}$$

where $F(E \geq \theta)$ is the fraction of *all* sequences (say, given a particular length L) that perform the function at a level at least θ.

Such a definition of functional information makes eminent sense (it implies that functional sequences are exponentially rare) and accurately links information content to functionality. For example, in the evolution of RNA molecules that encode an enzymatic activity [3], functional information is correlated to both activity and structural complexity. In the evolution of HIV protease [4], functional information can be seen to decrease when anti-viral drugs are given (reflecting the reduced activity of the protease), then to increase again as anti-viral resistance emerges. In fact, functional information defined in this manner turns out to be just the coarse-grained Shannon information content of the sequences [5].

A shortcoming of this measure is that only large groups of sequences can be characterized in this manner, because accurately estimating the fraction $F(E \geq \theta)$ necessitates large ensembles of functional sequences. Here we develop an approach that allows us to attach a score to *every individual sequence* in an ensemble, which then allows us to predict the level of functionality of individual sequences *not* in the ensemble. In particular, this approach allows us to study how functional information emerges from multi-site correlations within the sequence.

2. Information content of functional symbolic sequences

We now define the information content of a set of sequences $\boldsymbol{s}^{(k)}$ of length L that form an ensemble $\mathcal{S}$, given a particular environment within which they are functional [6–8]. The sequences $\boldsymbol{s}^{(k)}$ are written in terms of the state of a joint random variable $S = S_1, S_2, \ldots, S_L$ (one for each monomer), that is, $\boldsymbol{s}^{(k)} = (s_1^{(k)}, s_2^{(k)}, \ldots, s_L^{(k)})$, while the environment can be thought of as the state of an environment variable, that is, $E = e$. To define the information content of sequences, we need to define another ensemble: the ensemble of functional sequences $\mathcal{S}_e$, which is constrained by requiring function within environment e. Typically, the unconstrained ensemble is all the possible sequences of a given length L, while the ensemble $\mathcal{S}_e$ consists of only those sequences that are functional in e at a specified level. With this notation, the information content can be written as [6–8]

$$I(S : E = e) = H(S) - H(S|e), \tag{2.1}$$

where $H(S|e)$ is the entropy of sequences in $\mathcal{S}_e$: an entropy conditioned on the environmental variable taking on the state $E = e$. Further, $H(S)$ is the Shannon entropy of the unconstrained ensemble, which, if sequences of length L are drawn from an alphabet of D symbols, is simply $H(S) = L \log D$. The entropy of the constrained ensemble requires the knowledge of the likelihood p_k to find a functional sequence $\boldsymbol{s}^{(k)}$ within the constrained ensemble. Thus, equation (2.1), being the difference between the uncertainty about a non-functional ensemble and the uncertainty of the functional ensemble, simply reflects the knowledge of the function that constrains the ensemble: it is information about how to carry out the function.

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Given the likelihood p_k to find functional sequences of type k in $\mathcal{S}_e$, we can calculate the Shannon entropy of sequences in $\mathcal{S}_e$ as

$$H(S|e) = -\sum_{k}^{N_e} p_k \log p_k, \tag{2.2}$$

where N_e is the number of different functional sequences within $\mathcal{S}_e$. If we coarse-grain entropy (2.2) by assuming that each sequence appears with equal probability, then $p_k = 1/N_e$ and

$$I(S:e) = \log D^L - \log N_e = -\log\left(\frac{N_e}{D^L}\right), \tag{2.3}$$

which is Szostak's functional information since $F = N_e/D^L$.

Because most functional sequences of interest are exponentially rare, finding a sufficient number of them to accurately estimate the fraction $F(E \geq \theta) = N_e/D^L$ is difficult or nearly impossible. For example, while tens of thousands of variants of the 99-mer HIV-1 protease are known, they are not obtained by randomly sampling the 99-mer space, and therefore cannot be used to calculate functional information in this manner. However, it is possible to relate the information $I(S:e)$ to monomer, dimer and tri-mer entropies (and so on) as follows.

In general, a multivariable entropy such as $H(S|e) = H(S_1 S_2 \cdots S_L)$ can be decomposed into multi-mer correlation entropies using Fano's entropy and information decomposition theorems [9]). Specifically[1] (see appendix A for details)

$$H(S_1 \cdots S_L) = \sum_{m=1}^{L} (-1)^{m-1} \sum_{\sigma(1)\cdots\sigma(m)}^{L} I(S_{\sigma(1)} : \cdots : S_{\sigma(m)}), \tag{2.4}$$

and (the shared entropy $I(S_1 : \cdots : S_L)$ is also sometimes referred to as the *interaction information* of order L)

$$I(S_1 : \cdots : S_L) = \sum_{m=1}^{L} (-1)^{m-1} \sum_{\sigma(1)\cdots\sigma(m)}^{L} H(S_{\sigma(1)} \cdots S_{\sigma(m)}). \tag{2.5}$$

In these formulae, $\sigma(i)$ stands for a permutation of the index i. For example, the entropy decomposition theorem (2.4) can also be written as

$$H(S_1 \cdots S_L) = \sum_{i=1}^{L} H(S_i) - \sum_{i<j}^{L} I(S_i : S_j) + \sum_{i<j<k}^{L} I(S_i : S_j : S_k) - \cdots (-1)^{L-1} I(S_1 : \cdots : S_L). \tag{2.6}$$

While the decomposition theorems are general, the individual interaction information terms have also been introduced in terms of 'connected information' [15] (see also appendix B).

The decomposition theorems allow us to approximate the exact information content $I(S:e)$. To start with, if we neglect all correlations between variables, then the first-order approximation to $H(S|e)$ is

$$H_1(S|e) = \sum_{i=1}^{L} H(S_i), \tag{2.7}$$

where $H(S_i)$ is the (functional) monomer entropy of site i, that is,

$$H(S_i) = -\sum_{s_i} p_i(s_i) \log p_i(s_i), \tag{2.8}$$

with $p_i(s_i)$ the likelihood to find symbol s_i (one out of D possible ones) at position i.

[1]Fano's decomposition theorems are just specific information-theoretic examples of the more general inclusion-exclusion theorem that has been known for some time. For example, Sylvester [10] refers to it as a 'théorème logique bien connu'. As a consequence, these theorems keep getting reinvented [11–14].

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If we keep correlations between pairs of sites but neglect correlations between three sites, we obtain the second-order approximation of the entropy as

$$H_2(S|e) = \sum_{i=1}^{L} H(S_i) - \sum_{i<j}^{L} I(S_i : S_j), \tag{2.9}$$

where $I(S_i : S_j)$ is the entropy shared between variables S_i and S_j. We can deduce how this is written in terms of marginal entropies (entropies obtained by summing over one or more of the monomers) using the decomposition theorem (2.5) for $L = 2$ as

$$I(S_i : S_j) = H(S_i) + H(S_j) - H(S_i S_j), \tag{2.10}$$

the standard Shannon formula for information.

Because $H(S_i)$ and $I(S_i : S_j)$ can be estimated from finite ensembles of functional sequences, approximations such as (2.9) are sufficient to estimate the information content $I(S : e)$ as long as correlations between three or more variables can be neglected (e.g. [4]). As long as higher-order multivariate correlations are weak, higher-order approximations to the sequence entropy $H(S|e)$ should converge, meaning that approximations to the information content taking into account increasingly higher correlations should be increasingly accurate.

While these approximations are useful for calculating the information content of sequences from finite ensembles, they do not allow us to correlate *individual* characters of a sequence (for example, its particular level of functionality) to information because, as we have emphasized repeatedly, information is not a property of a single sequence. However, each sequence is also a *pattern*, so it should be possible in principle to relate the individual pattern in the sequence to its function. Indeed, methods of this sort have been developed before: they use sophisticated techniques to link the patterns to function, either by regression [16,17] or by using a multiple-sequence alignment (MSA) to train a probabilistic model (usually called 'Potts' or 'Ising' model) [18–21].

In the following section, we outline an approach that allows us to link patterns to function *directly*, without regression or fitting a Potts model. The approach is based on the entropy and information decompositions (2.4) and (2.5) outlined above. We first review how the linkage between pattern and function is achieved in the standard approaches (i.e. regression or fitting to a Potts model), and then show that the information decomposition formulas render regression or fitting is unnecessary. We review the concept of the position weight matrix, and show that functional scores for sequences can be defined based on those matrices in a model-free manner.

3. Model-free linkage of pattern to function

A common technique to model correlations between variables in a sequence $\boldsymbol{s} = (s_1, s_2, \ldots, s_L)$ is the so-called Potts or Ising model Hamiltonian (energy functional)[2] that can be written as [18,19]

$$E(\boldsymbol{s}) = \sum_{i=1}^{L} h_i(s_i) + \sum_{i<j}^{L} J_{ij}(s_i, s_j). \tag{3.1}$$

Here as before, the s_i are the sequence monomers taken from an alphabet of size D. For neuronal firings, the s_i are binary, while for much of the literature on molecules, the s_i are reduced to a small

[2]Technically speaking, the model is neither a Potts nor an Ising model (these are well-known models in condensed matter physics that describe interactions between spins), as those models only have nearest-neighbour interactions. It is best called an infinite-range Ising-like model.

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alphabet, for example, $s_i = 0$ for wild-type monomer, and $s_i = 1$ for a mutant. The $L \cdot D$ parameters $h_i(s_i)$ and $\binom{L}{2}D^2$ parameters $J_{ij}(s_i, s_j)$ are Lagrange multipliers obtained by maximizing the entropy

$$H(\boldsymbol{s}) = -\sum_{k=1}^{D^L} p_k \log p_k, \tag{3.2}$$

constrained to reproduce the monomer and dimer probabilities $p(s_i)$ and $p(s_i, s_j)$ that can be obtained from the MSA. The probability $p_k = P(\boldsymbol{s}^{(k)})$ is incidentally the same as defined in (2.2): it is the likelihood to find sequence $\boldsymbol{s}^{(k)}$ by chance. Maximization under the constraints yields the familiar Boltzmann distribution

$$p_k = \frac{1}{Z}\,\mathrm{e}^{-E(\boldsymbol{s}^{(k)})}, \tag{3.3}$$

with $E(\boldsymbol{s})$ from (3.1) above, and where

$$Z = \sum_k \mathrm{e}^{-E(\boldsymbol{s}^{(k)})} \tag{3.4}$$

is the partition function.

Now, let us take a closer look at the entropy $H(\boldsymbol{s})$. Using (3.3), we can write this as

$$H(\boldsymbol{s}) = \log Z + \sum_{i=1}^{L}\sum_{s_i} h(s_i)p_i(s_i) + \sum_{i<j}^{L}\sum_{s_i,s_j} p_{ij}(s_i, s_j)J_{ij}(s_i, s_j), \tag{3.5}$$

where $p_i(s_i)$ is the probability to find symbol s_i at site i, and $p_{ij}(s_i, s_j)$ is the probability to find the pair $s_i s_j$ at sites i and j. These probabilities are obtained from (3.3) by summing over those sites that are not i or j, or both, respectively.

It is clear that the right-hand side of equation (3.5) is an approximation of the 'true' entropy $H(\boldsymbol{s})$, fitted to the MSA using the first- and second-order Lagrange multipliers. Higher-order terms are neglected. In the Potts model approach, the term $\log Z$ on the right-hand side of (3.5) is neglected, but we now show that it can be calculated instead.

It turns out that we can approximate the left-hand side using averages taken directly from the MSA, up to any order we want. This is possible because we can decompose $H(\boldsymbol{s})$ exactly using the entropy decomposition theorem, for example, written in the form (2.6). Using this approximation, we will be able to determine $\log Z$, and eliminate the Lagrange multipliers. As a consequence, no fitting of these parameters to data is necessary.

Because the Potts model parameters only consider up to bi-partite correlations, we will neglect in the following terms of order three or higher in (2.6) for now, but it is clear that we could keep them as long as we fit an appropriate third-order term in the Hamiltonian (3.1). Using $p_i(s_i)$ directly from the MSA, along with the $p_{ij}(s_i, s_j)$, and so forth, we can insert those terms in (3.5) to yield

$$\begin{aligned}\log Z &= -\sum_i\sum_{s_i} p_i(s_i)\Big(\log p_i(s_i) + h(s_i)\Big) + \sum_{i<j}\sum_{s_i,s_j} p_{ij}(s_i, s_j)\left(\log\frac{p_i(s_i)p_j(s_j)}{p_{ij}(s_i, s_j)} - J_{ij}\right)\\ &= \sum_i\Big(H(i) - \langle h_i\rangle\Big) - \sum_{i<j}\Big(H(i:j) + \langle J_{ij}\rangle\Big).\end{aligned} \tag{3.6}$$

In (3.6), we defined the average Lagrange coefficients

$$\langle h_i\rangle = \sum_{s_i} p_i(s_i)h(s_i) \tag{3.7}$$

and

$$\langle J_{ij}\rangle = \sum_{s_i,s_j} p_{ij}(s_i, s_j)J_{ij}(s_i, s_j). \tag{3.8}$$

We also introduced the simplified notation $H(i) \equiv H(S_i)$ and $H(i:j) \equiv H(S_i : S_j)$ and so on, which we will use from now on.

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In order to understand the term $\log Z$, we turn to Berg–von Hippel (BvH) theory [22,23]. Berg and von Hippel developed a framework that allowed them to assign scores to putative DNA binding sites that are linked to the likelihood that such a putative binding site is functional. The framework is a maximum entropy approach just like the Potts model approach from [19], except that Berg and von Hippel assumed that every single nucleotide in the binding site is at maximum entropy. The score that each sequence is assigned is an *energy score*, where the consensus sequence of the MSA for that site has zero energy (the 'ground state'). In fact, the energy score of a DNA binding site is given by an energy functional that is precisely of the form (3.1), but without the second-order term (as correlations between sites are considered not to be important in DNA binding).

Unlike in the Potts-model approach, the counterparts to $h_i(s_i)$ are *not* fitted to the MSA in the BvH approach. Rather, they are directly extracted from the MSA in terms of a matrix as[3]

$$M^{(1)}_{ia} = \lambda \log \frac{p_0(s_i)}{p_i(s_a)}, \tag{3.9}$$

because this is the expression that maximizes the single-site entropy (subject to constraints). We can see this by writing

$$p_i(s_a) = p_0(s_i)\, \mathrm{e}^{-(M^{(1)}_{ia}/\lambda)}. \tag{3.10}$$

In equation (3.9), λ is a Lagrange multiplier that allows us to set the scale of the energy just as we did in the earlier maximization (we set $\lambda = 1$ here). Furthermore, $p_0(s_i)$ is the probability (likelihood) to find the consensus monomer at site i. The consensus monomer is the one that has the highest frequency in the MSA at that position, that is, $p_0(s_i)$ is the largest among the $p_i(s_a)$. This assures that the consensus sequence is the one with vanishing energy. The matrix $M^{(1)}$ is called the 'position-weight matrix' (PWM) of a sequence [24], and it can be used to score any putative nucleotide sequence. For example, using the PWM (3.9) obtained from an alignment of functional DNA binding sequences, we can define the (first-order) energy score of an *arbitrary* sequence $\boldsymbol{s}^{(k)}$ as

$$E_1(\boldsymbol{s}^{(k)}) = \mathrm{Tr}(M^{(1)} G_k), \tag{3.11}$$

where G_k is the $D \times L$ matrix representing sequence $\boldsymbol{s}^{(k)}$, which has a '1' at position i if symbol a is found there, and a '0' otherwise, and Tr is the matrix trace. With this definition, the consensus sequence $\boldsymbol{s}_0$ has $E_1(\boldsymbol{s}_0) = 0$, and sequences that carry symbols at that position that rarely occur in the MSA score badly (have high energy). For cases where the symbol does not occur at all in the MSA at that position, we have to introduce the concept of pseudocounts in order to deal with zeros in the denominator of (3.9), as we discuss in more detail below. Using a threshold where all sequences with an energy score lower than the threshold are predicted to be functional DNA binding sites, BvH theory is fairly successful at predicting DNA binding sites in a genome (e.g. [25]), and even at detecting correlations between binding sites [26].

We have now seen that BvH theory can assign a score that reflects the functionality of a sequence pattern without the need to fit any parameters. But in order to quantify the functionality of sequences with significant correlations between sites, we have to go beyond that theory. We now introduce a *second-order* PWM (as BvH in fact did in the appendix to [22], but for adjacent monomers only)

$$M^{(2)}_{ijab} = \log\left(\frac{p_0(s_{ij})}{p_{ij}(s_a, s_b)}\right). \tag{3.12}$$

Just as before, $p_{ij}(s_a, s_b)$ is the likelihood to find symbol combination s_a, s_b at the pair of sites i, j, while $p_0(s_{ij})$ is the likelihood to find the consensus pair there.

Defining this matrix implicitly assumes that beyond monomers, any *pair* of monomers is in statistical equilibrium as well. While assuming that even higher-order correlations are in statistical

[3]We changed notation slightly in equation (3.9), from $p_i(s_i)$ in equation (2.8) to $p_i(s_a)$ here, to conform to BvH theory. The difference is that in BvH theory it is convenient to index the symbol (a runs from 1 to 4, indexing the four different nucleotides), so that it is obvious that $p_i(s_a)$ is a matrix element. When writing the probability in the form $p_i(s_i)$, rather than summing over an index we sum over the symbols s_i instead.

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equilibrium might be increasingly doubtful, it is certainly more likely than assuming that the entire sequence is at maximum entropy, as (3.3) does.

We can now use BvH theory to calculate $\log Z$ and insert it into equation (3.6). In standard BvH theory, $\log Z$ is given by (recall we are setting $\lambda = 1$ here)

$$\log Z = -\sum_{i=1}^{L} \log p_0(s_i), \tag{3.13}$$

but this is only the first-order approximation. In general, for any sequence $\boldsymbol{s}$ we must have

$$Z = \frac{1}{p_0(s_1 \cdots s_L)}, \tag{3.14}$$

which we can expand so that[4]

$$-\log Z = \sum_{i=1}^{L} \log p_0(s_i) - \sum_{i<j}^{L} \log \left[\frac{p_0(s_{ij})}{p_0(s_i)p_0(s_j)} \right] + \cdots . \tag{3.15}$$

Inserting this into (3.6) we can now identify

$$\langle h_i \rangle = H(i) + \log p_0(s_i) \tag{3.16}$$

and

$$\langle J_{ij} \rangle = -H(i:j) - \log \left[\frac{p_0(s_{ij})}{p_0(s_i)p_0(s_j)} \right], \tag{3.17}$$

that is, the averages of the Lagrange parameters are in fact given by averages that can be extracted from the MSA directly. Equations (3.16) and (3.17) imply that there is a relationship between the energy contribution to a site $\langle h_i \rangle$ and the entropy $H(i)$ of that site, and it is precisely the relationship that is summarized by the thermodynamical Helmholtz free-energy relation

$$\Omega = E - TS, \tag{3.18}$$

where $\Omega = -\log Z$ is the free energy, E is the average energy and S is the Gibbs entropy (T, of course, is the temperature). We will see below that not only are the averages related in this manner, but the parameters of the Potts model are given entirely by quantities that can be extracted directly from the MSA, without the need of any optimization, namely:

$$h_i(s_a) = M_{ia}^{(1)} \tag{3.19}$$

and

$$J_{ij}(s_a, s_b) = M_{ijab}^{(2)}. \tag{3.20}$$

Thus, in this approach the energy of a particular sequence $E(\boldsymbol{s})$ can now be calculated from the first- and second-order PWMs in a model-free manner.

In the next section, we extend this approach to take into account arbitrary-order correlations between monomers, in order to construct predictive energy and information scores for arbitrary sequences. After that, we will use these scores to predict the functionality of sequences that have strong correlations between sites, and are drawn from a high-dimension alphabet.

4. Information and energy scores

In the previous section, we showed that the Potts-model approach creates an energy score (3.1) by fitting Lagrange parameters to data, even though such a fitting procedure is completely unnecessary: when evaluating the term $\log Z$, averages over the Lagrange parameters are given by single-site and pairwise entropies. But given that the BvH model was able to define scores for

[4]This is the expansion of the joint probability that is at the root of Fano's entropy and information decomposition theorems (2.4) and (2.5), see appendix A.

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each sequence (which, when averaged over, produces entropies), could we generate energy and information scores *directly* from the position weight matrices *without fitting any parameters*?

In what follows, we will write down energy functionals (scores) based on the higher-order (model-free) PWMs, and then show how those are related to *information scores*. In the section following, we will use these scores to construct *predictors* that can be used to predict the functionality of sequences based on the sequence pattern only. The reason why we need to construct predictors rather than use the energy and information scores directly will become clear when we apply them to actual functional data with strong correlations.

Rewriting the first-order energy score (3.11) of a sequence $\boldsymbol{s}$ as

$$\langle E_1\rangle_e = H_1(S|e) + \sum_i^L \log p_0(s_i), \tag{4.1}$$

where $H_1(S|e)$ is the first-order approximation to the Shannon entropy of the functional ensemble $\mathcal{S}_e$, defined in (2.7), we recognize it as the first-order version of the Helmholtz free energy relationship from thermodynamics. Given that the average energy can be written as the average of an energy score, is there an information score whose average is the Shannon information of a sequence ensemble? It turns out that there is.

We can introduce the first-order information matrix[5]

$$R_{ia}^{(1)} = \log\left[\frac{p_i(s_a)}{1/D}\right], \tag{4.2}$$

defined in such a way that its average is the first-order Shannon information (if we take logs to the base of the alphabet D)

$$\langle R_1\rangle_e = \sum_{i,a} p_i(s_a) R_{ia}^{(1)} = L - \sum_i^L H(i), \tag{4.3}$$

where

$$H(i) = -\sum_a p_i(s_a)\log p_i(s_a). \tag{4.4}$$

Note that this information matrix is defined in close analogy to the position weight matrix (3.9), except that rather than comparing $p_i(s_a)$ to the likelihood of the most common symbol, the information matrix is constructed by comparing $p_i(s_a)$ to the unbiased expectation, that is, $1/D$.

The information score for a sequence $\boldsymbol{s}_k$ constructed using the first-order information matrix is constructed just like (3.11)

$$R_1(\boldsymbol{s}^{(k)}) = \mathrm{Tr}(R^{(1)}G_k). \tag{4.5}$$

We now construct both energy and information scores that include higher-order corrections. Using $E(s_1\cdots s_L) = \log(p_0(s_1\cdots s_L)/p(s_1\cdots s_L))$ and the expansion of the joint probability (see appendix A) we can write

$$E(s_1\cdots s_L) = \sum_{i=1}^L E(s_i) - \sum_{i<j}^L E(s_i:s_j) + \sum_{i<j<k}^L E(s_i:s_j:s_k) - \cdots + (-1)^{L-1}E(s_i:s_j:\cdots s_L). \tag{4.6}$$

Here, we defined joint and *shared* energy scores as

$$E(s_i) = \log\frac{p_0(s_i)}{p_i}, \tag{4.7}$$

$$E(s_i s_j) = \log\frac{p_0(s_{ij})}{p_{ij}} \tag{4.8}$$

and

$$E(s_i:s_j) = E(s_i) + E(s_j) - E(s_i s_j). \tag{4.9}$$

[5]This is in fact Schneider's information matrix introduced in [27], but with a uniform prior $1/D$ and without finite sample size correction, which we will take care of using pseudocounts instead.

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Analogous relations hold for information scores for monomers, dimers and so on. Note that expression (4.6) exactly mirrors the entropy decomposition (2.6), but it holds for every sequence, not just for the average over sequences.

We now construct approximations to the exact energy score (4.6). The first-order approximation was already given in (3.11), but we rewrite it here:

$$E_1(s_1 \cdots s_L) = \sum_{i=1}^{L} E(s_i), \tag{4.10}$$

$$E_2(s_1 \cdots s_L) = \sum_{i=1}^{L} E(s_i) - \sum_{i<j}^{L} E(s_i : s_j) \tag{4.11}$$

and

$$E_3(s_1 \cdots s_L) = \sum_{i=1}^{L} E(s_i) - \sum_{i<j}^{L} E(s_i : s_j) + \sum_{i<j<k}^{L} E(s_i : s_j : s_k). \tag{4.12}$$

We can rewrite the correction terms as a function of joint (rather than shared) energy scores using the decomposition theorem. For example, using (4.9), we find

$$\begin{aligned} E_2(s_1 \cdots s_L) &= \sum_{i=1}^{L} E(s_i) - \sum_{i<j}^{L} (E(s_i) + E(s_j) - E(s_i s_j)) \\ &= \sum_{i<j}^{L} E(s_i s_j) - (L-2) \sum_{i=1}^{L} E(s_i). \end{aligned} \tag{4.13}$$

Likewise, if we neglect the fourth-order correlations

$$E_3(s_1 \cdots s_L) = \sum_{i<j<k}^{L} E(s_i s_j s_k) - (L-3) \sum_{i<j}^{L} E(s_i s_j) + \frac{1}{2}(L-2)(L-3) \sum_{i=1}^{L} E(s_i). \tag{4.14}$$

Turning now to information scores (here, $R(s_i) = \log[Dp_i(s_i)]$, $R(s_i s_j) = \log[D^2 p_{ij}(s_i s_j)]$ and so on, and logs are to the base D) we can define

$$I_1(s_1 \cdots s_L) = \sum_{i=1}^{L} R(s_i). \tag{4.15}$$

$$I_2(s_1 \cdots s_L) = \sum_{i=1}^{L} R(s_i) - \sum_{i<j}^{L} R(s_i : s_j) \tag{4.16}$$

and

$$I_3(s_1 \cdots s_L) = \sum_{i=1}^{L} R(s_i) - \sum_{i<j}^{L} R(s_i : s_j) + \sum_{i<j<k}^{L} R(s_i : s_j : s_k), \tag{4.17}$$

where the shared information scores $R(s_i : s_j)$ etc. are defined analogously to the shared energy scores. Averaging these over the sequences in the functional ensemble $\mathcal{S}_e$ will return the approximations (2.7) and (2.9), and so on. Since these sequence functionals are constructed so that they reflect function (low-energy and high-information scores for highly functional sequences) they can in principle be used to classify sequences into functional versus non-functional (at a given level of functionality). We expect that the accuracy of a discriminator based on these functionals improves if higher-order correlations are included in the score function.

5. Application to self-replicating sequences

In the previous section, we saw that energy and information scores are built up hierarchically, adding pairwise, triplet and higher correlations until the full functionality emerges. But what level of correlations do we need to keep track of in order to accurately capture a sequence's

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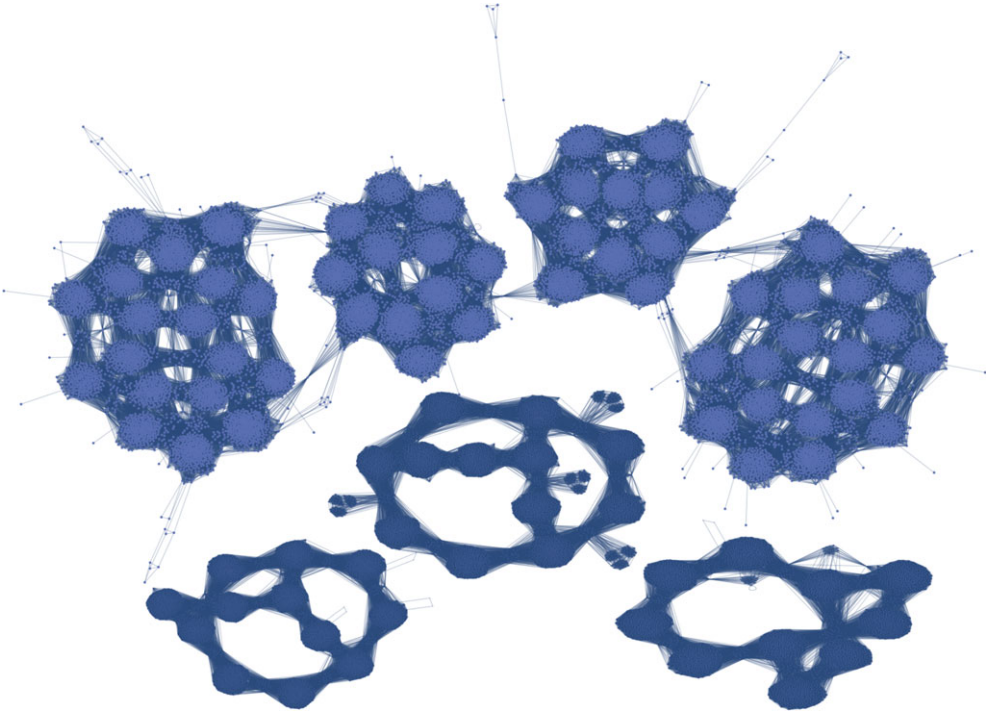

Figure 1. The four largest connected components of the network of self-replicators comprise 31 592 sequences (about 87% of the total). In these networks, two replicators are joined by an edge if they differ by exactly one point-mutation. (Online version in colour.)

functionality? To test this, we use a dataset of sequences in which correlations are important, but is also large enough so that higher-order PWMs are statistically significant. Previously, we analysed a set of 36 171 sequences of self-replicating computer programs from the digital life system Avida (e.g. [28–30]). Avida simulates a computer within a standard computer, within which self-replicating programs written in a custom programming language live and evolve. The programming language typically uses only 26 instructions (conveniently labelled by the lowercase letters a–z) that are chosen in such a way that programs are highly evolvable. Because the replication of instructions is noisy (giving rise to mutations), populations of programs evolve and adapt to user-specified fitness landscapes. The $N_e = 36\,171$ sequences that we analyse here represent the complete set of all possible self-replicating programs of length $L = 9$ (out of a possible $26^9 \approx 5.43 \times 10^{12}$) obtained via an exhaustive search [31]. 'Self-replication' is defined here as the capacity to form a growing colony of programs (in the absence of mutations) and does not necessarily require perfect copying of information. Understanding the structure of the fitness landscape of self-replicators is an important element in trying to understand how replication could have emerged from non-replication, i.e. the origin of life [5,32].

The 36 171 sequences form complex clusters in genotype space (figure 1). We can immediately calculate the information content of these 9-mers using Szostak's formula (1.1), which gives[6]

$$I_9 = \log_{26} \frac{36\,171}{26^9} \approx 5.77 \text{ mers}, \tag{5.1}$$

that is, about 64% of the sequence is information. The smallest possible replicator in this world has $L = 8$, with an even more compressed code [31].

[6]Choosing a base for the logarithm gives units to entropy and information. In the following, we will take logs to the base of the alphabet, giving units of 'mers'. An n-mer has n mers of potential information (entropy).

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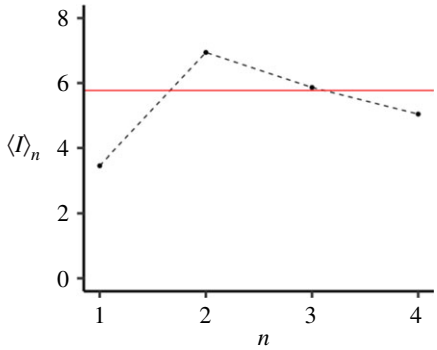

Figure 2. Mean information scores (Shannon information) as a function of the degree of correlations taken into account. $\langle I \rangle_1$ corresponds to the information disregarding pairwise correlations between sites, while $\langle I \rangle_2$ disregards third-order correlations, and so on. The solid horizontal line is the exact information $I_9 = 5.77$ mers. (Online version in colour.)

(a) Emergence of information from multivariate correlations

Earlier we saw that we can approximate the total Shannon information stored in a sequence using the information decomposition theorem. Applying this to the $\mathcal{S}_e$ of 9-mer replicators $\boldsymbol{s}_9$, we find

$$\begin{aligned} I_9 &= 9 - H(\boldsymbol{s}_9) \\ &= 9 - \sum_{i=1}^{9} H(i) + \sum_{i<j}^{9} I(i:j) - \sum_{i<j<k}^{9} I(i:j:k) \cdots - I(S_1 : \cdots : S_9). \end{aligned} \tag{5.2}$$

We can now study how this information emerges from correlations, by averaging the information scores (4.15) over $\mathcal{S}_e$ and thus recover the exact information (5.1) by taking into account successively higher-order terms.

In figure 2, we show the approximations to the exact information (5.1) as a function of the order n of the approximation. Estimating the information content from single-site abundances severely underestimates the true information. Taking into account only second-order correlations *overestimates* the true information, however, because the third-order term is of opposite sign. As expected from the information decomposition theorem, the approximations move closer to the exact value as more correlations are incorporated, suggesting that higher-order multivariate correlations become weaker, and that the sum converges.

(b) Prediction of function from information scores

In this section, we will test how much 'correlation information' (information stored in correlations between sites, as opposed to bias in per-site frequency) is necessary to distinguish functional from non-functional sequences. Because we will now be scoring sequences that are *not* in the set $\mathcal{S}_e$, we have to introduce pseudocounts in the PWMs. As is clear from (3.9), if a particular symbol s_a at position i does not occur in the MSA, then the maximum-likelihood estimator of the probability $p_i(s_a)$ vanishes, leading to an infinite matrix element. As a consequence, the score of a sequence that contains such a symbol at the position is not well defined. A common cure for this defect is to introduce pseudocounts, which add a constant to every possible symbol count. While the common Laplace pseudocount adds 1 to each count ($n_a \to n_a + 1$ and therefore $p_i(s_a) \to (n_a + 1)/(N_e + D)$), the pseudocount is often used as a variable that can be adapted to the statistics of the dataset in question (e.g. [33,34]). Indeed, if the ensemble of functional sequences is small, then it is imperative to use such pseudocounts in order to correct for a finite-size bias in entropy estimation [33]. In what follows, we adopt a pseudocount of 10^{-6} throughout, but note that the results on classification accuracy scale extremely weakly over the 12 orders of magnitude

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of pseudocounts we tested. Generally speaking, an infinitesimal pseudocount moves information scores of unrepresented symbols far out into the negative, and energy scores far to the positive. The pseudocount for higher-order PWMs (we construct those explicitly up to $n = 4$, where n is the order of correlations that are included) is adjusted so that the marginal probabilities remain correct.

In order to predict the function of an arbitrary sequence in light of an MSA of $N \leq N_e$ functional ones, we need to construct a classifier in such a manner that, given a particular threshold τ, the score divides functional from non-functional sequences. For energy scores, we would deem those sequences with a score $E(\boldsymbol{s}) < \tau$ as functional, while for information scores $I(\boldsymbol{s}) > \tau$ predicts function. Note that because we have exhaustively classified this dataset already, we can determine the accuracy of classification (given a subset of sequences $N < N_e$ for a given classifier) perfectly.

In principle, we can construct classifiers directly from the information score functions (4.15), but because the correction terms to I_1 involve the summation of many terms with alternating signs, the pseudocount prevents these classifiers from being effective as they can introduce spurious correlations.[7] While it is possible to make sure no spurious correlations are introduced for any chosen level, it appears to be impossible to construct pseudocounts in a manner that avoids spurious correlations at *all* levels of n. However, it turns out that sums over the multiple-site energy and information functionals $E(s_i s_j \cdots)$ and $R(s_i s_j \cdots)$ do work well as classifiers, because they do not rely on cancellations. For example, we can define the following information classifiers (similar classifiers can be built from energy functionals):

$$D_1(\boldsymbol{s}) = \sum_{i=1}^{L} R(s_i), \tag{5.3}$$

$$D_2(\boldsymbol{s}) = \sum_{i<j}^{L} R(s_i s_j), \tag{5.4}$$

$$D_3(\boldsymbol{s}) = \sum_{i<j<k}^{L} R(s_i s_j s_k) \tag{5.5}$$

and

$$D_4(\boldsymbol{s}) = \sum_{i<j<k<m}^{L} R(s_i s_j s_k s_m). \tag{5.6}$$

The classifier $D_1(\boldsymbol{s})$ assigns an information score to the sequence $\boldsymbol{s}$, but ignores pairwise or higher-order correlations. 'Typical' sequences (meaning sequences like those in the MSA) have a high information score, whereas random (atypical) sequences have low scores (often negative, as information scores are not strictly positive or zero). Classifier D_2 goes beyond D_1 and takes into account pairwise correlations, while D_3 takes third-order as well as second-order correlations into account. Classifier D_4 ought to be the most accurate, as it takes into account correlations up to order 4.

To test the quality of the classifiers, we create test data containing 10^5 sequences. The test data are sequences of length 9 taken from the same alphabet, but which are *not* within the MSA created from $\mathcal{S}_e$. Specifically, we create three ensembles: $\mathcal{S}_R$ with randomly generated sequences of $L = 9$ that we made sure are all non-functional, as well as a set of one-mutants $\mathcal{S}_1$ and two-mutants $\mathcal{S}_2$ that are one or two mutations away from functional sequences, but tested to ensure they are not functional. We expect the one-mutant set in particular to be the hardest to classify.

We show typical density distributions of information scores for these classifiers in figure 3. These classifiers were all 'trained' on the full set $\mathcal{S}_e$, meaning the PWMs $M^{(n)}$ were created with the entire set of replicators in the MSA. As expected, the set of random sequences has a density distribution (purple in figure 3*a–d*) far removed from the functional sequences (light blue), but the one-mutant and two-mutant sequences have a significant overlap with functional sequences if D_1

[7]This holds equally when using the energy score functions (4.10) as classifiers.

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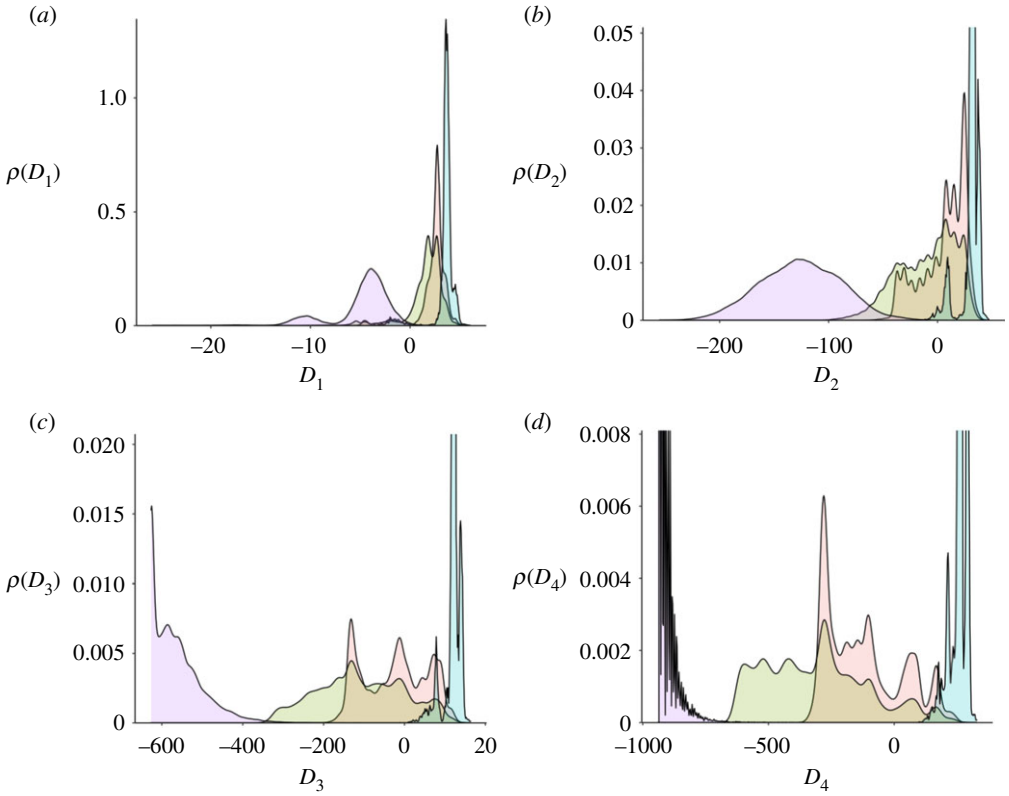

Figure 3. Density plots of scores generated by the classifiers (5.3)–(5.6), with PWMs using all sequences in $\mathcal{S}_e$ and a pseudocount 10^{-6}. Note that for visibility, some distributions are cropped at the top. (*a*) Density of scores obtained with D_1 for true sequences ($\mathcal{S}_e$, light blue), false sequences ($\mathcal{S}_R$, purple), single-mutant false ($\mathcal{S}_1$, red) and double-mutant false ($\mathcal{S}_2$, green). (*b*) Density of scores for classifier D_2. Colours as in (*a*). (*c*) Density of scores for classifier D_3. (*d*) Density of scores for classifier D_4. (Online version in colour.)

is used as a classifier. This overlap leads to false positives (non-functional sequences classified as functional), which the more complex classifiers should be able to avoid. Including higher-order correlations indeed reduces this overlap, until the overlap (misclassification) almost completely disappears in D_4, even between functional sequences and their non-functional one-mutants (all the functional one-mutants are contained in $\mathcal{S}_e$, of course).

We can characterize the accuracy of the classifier as a function of the size of the training set by measuring the quality score of a receiver operating characteristic (ROC, e.g. [35]) plot. ROCs are generated by plotting the false positive rate (FPR) against the true positive rate (TPR) as a function of the threshold τ in the classifier. A typical ROC (for classifier D_2 trained on 1% of $\mathcal{S}_e$ used on 100 000 non-functional single-point mutants of the functional set) is shown in figure 4. The quality score Q of the classifier is obtained by calculating the area under the curve (AUC) of the ROC. In comparison, a random classifier where the TPR increases linearly with the FPR has AUC = 0.5.

It is clear from figure 5 that all classifiers work well even when only 0.1% of the true replicators (36 sequences) are used to create the MSA, but for training sets that small, the classifiers D_3 and D_4 that use three-partite and four-partite correlations perform less well than D_1 and D_2 on the test data for one- and two-mutants of functional sequences, which can be attributed to overfitting. Indeed, for datasets this small, the equilibrium assumption behind the max entropy approach will be violated for those PWMs. However, for larger MSAs, including higher-order correlations significantly improves the performance of the classifier. When using the full set of replicators,

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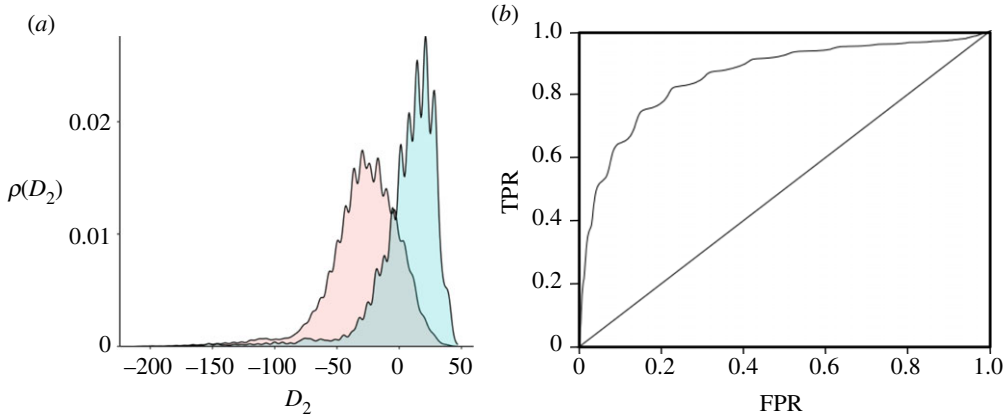

Figure 4. (*a*) Distribution of functional sequences $\mathcal{S}_e$ (blue) and of 100 000 uniformly sampled non-functional single-point mutants of functional sequences ($\mathcal{S}_1$, red), both evaluated by a classifier D_2 constructed from 1% of the N_e sequences in $\mathcal{S}_e$, as a function of the score. (*b*) ROC plot for the classifier D_2 in (*a*) as the threshold parameter τ is varied from $\tau = 45$ to $\tau = -211$. True positives are functional. The diagonal line is the ROC of a random (non-specific) classifier. (Online version in colour.)

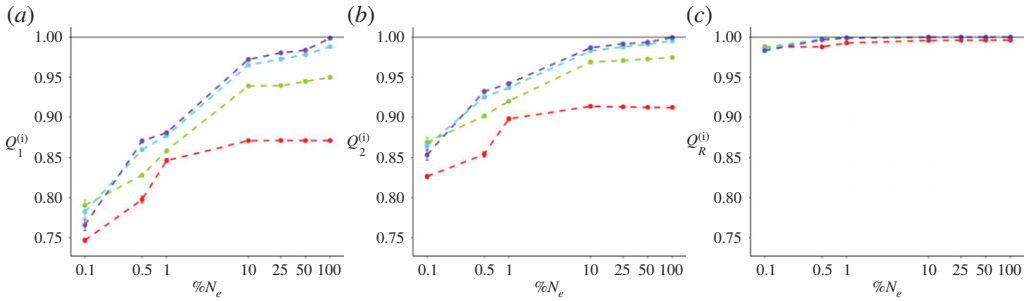

Figure 5. Quality Q of classifiers (AUC of the ROC curves) when testing against different test sets, as a function of the training set size (in percentage of the full set of size N_e). Note log scale for % of N_e. In all panels, red points and lines refer to the first-order classifiers D_1, green dots and lines depict the performance of second-order classifiers D_2, while the third- and fourth-order classifiers are in cyan and purple, respectively. Error bars are standard error over 10 replicates (except for $N = N_e$). (*a*) Q_1 is the quality score of classifiers for a test set made from 100 000 non-replicating one-mutants of the functional set $\mathcal{S}_1$. (*b*) Q_2 quantifies the quality of the classifiers when tested on the two-mutant non-functional set $\mathcal{S}_2$. (*c*) Q_R measures the performance of the classifiers when tested on 100 000 randomly generated non-functional 9-mers. (Online version in colour.)

taking into account up to four-way correlations appears to capture almost all of the information encoded in the sequences. On random sequences, all classifiers perform close to perfectly even on the smallest training set size, and including multivariate correlations is clearly overkill. The quality of classifiers based on energy functionals is the same as for those described here (data not shown), as they are intimately connected via the Helmholtz free energy relation.

6. Discussion

Fano's entropy and information decomposition theorems allow us to understand how functional information emerges from the combination of multivariate correlations. Here we have shown that defining energy and information functionals built from multivariate position weight matrices allows us to capture the correlations inherent in multiple sequence alignments in a model-free manner. Standard approaches to fitting MSAs rely on computationally expensive inverse models to fit the pairwise correlations in the data, but often must reduce the alphabet size considerably

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to reduce the number of parameters in the Potts or Ising models, in particular in the analysis of protein data. It is clear from the present analysis that such a fit is entirely unnecessary, as the model's marginal probabilities can be extracted from the data as long as the maximum entropy assumption is used on single sites, pairs of sites, triples, and so on. It should be clear from the construction that the methods we have outlined apply to symbolic sequences of arbitrary alphabets, including (but not limited to) neuronal spike trains, nucleic acid sequences, or amino acid sequences.

We tested this approach on the largest computational genotype–phenotype map created to date. Having a complete genotype–phenotype map allows us to calculate the information content of functional sequences exactly, and study how this information is built up from lower-order correlations. Further, a complete dataset helps in creating ensembles that can be used to test the classification accuracy of function prediction. In particular, we used the model-free approach to construct classifiers that can distinguish self-replicating from non-self-replicating sequences even when the non-functional sequences are single-site mutations of the functional ones. Of course, this approach needs to be tested on biological datasets with significantly longer sequences. However, since the computational cost of creating these complex classifiers scales mainly with the level of multivariate correlations that are included, we expect to be able to classify sequences of several hundreds of sites with full alphabet size (for example, $D = 20$ for proteins) taking into account all tri-partite correlations. For binary data such as neuronal spike trains, we expect to be able to handle sequences of thousands of sites, possibly including even more than 4-neuron correlations.

Sequence models of the Ising or Potts-type perform markedly better than the standard Hidden Markov Models (HMMs) that have been developed and used since the early 1970s [36], because they can take into account arbitrary correlations between pairs of symbols, rather than just those between adjacent symbols in HMMs. However, going beyond pairwise correlations is deemed impossible for Ising models because of the associated explosion in the number of parameters. In exceptional cases (sparse networks of firing neurons), it is possible to move beyond the pairwise approximation [37,38] by estimating $\log Z$ using the likelihood of the most common firing motif (there, the totally quiescent state). The theory we have outlined here shows (see in particular equation (3.14)) why such an ad hoc approximation works.

While more general pattern classification tasks (of which sequence classification is but a subset) can nowadays be carried out by training deep neural networks on annotated data, these methods have also displayed severe drawbacks [39,40], for example by being easily fooled by deceptive patterns. Furthermore, training such models on a large corpus of data is computationally expensive. The model-free classifiers that we have described here are different from deep networks in important ways, as they do not require any training, and their vulnerability to overfitting is easily controlled by adjusting the level of correlation n to be included to the size of (number of sequences in) the MSA. Applications of this method to biological sequence data and neuronal data streams are forthcoming.

Data accessibility. The code to replicate the computations, along with the file of 36 171 replicators of length 9 in Avida, is available at https://figshare.com/articles/online_resource/Emergence_of_functional_information_from_multivariate_correlations/19064813.

Authors' contributions. C.A.: conceptualization, formal analysis, funding acquisition, methodology, project administration, supervision, writing—original draft, writing—review and editing; N.C.G.: data curation, formal analysis, investigation, methodology, software, validation, visualization, writing—review and editing.

Both authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest declaration. We declare we have no competing interests.

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Appendix A. Entropy and information decomposition theorems

Fano's entropy and information decomposition theorems are based on an expansion of joint and shared probability distributions. For a joint random variable $S = \otimes_{i=1}^{L} S_i$ with probability distribution $p(s_1 \cdots s_L)$ we can construct marginal distributions by summing over (tracing out) sites. For example, the single site distribution p_i is obtained as

$$p_i(s_i) = \sum_{s_j \neq s_i} p(s_1 \cdots s_i \cdots s_L), \tag{A 1}$$

while the joint distribution for sites i and j is obtained by tracing out all sites except i and j

$$p_{ij}(s_i, s_j) = \sum_{s_m \neq s_i \neq s_j} p(s_1 \cdots s_i \cdots s_j \cdots s_L). \tag{A 2}$$

The pairwise shared probability $p(i:j)$ is constructed from these marginals as

$$p(i:j) = \frac{p_{ij}(s_i, s_j)}{p_i(s_i) p_j(s_j)} \tag{A 3}$$

so that the pairwise Shannon information is then simply the average of $\log p(i:j)$ over the joint pairwise distribution p_{ij}

$$I(i:j) = -\sum_{s_i, s_j} p_{ij}(s_i, s_j) \log p(i:j) = H(S_i) + H(S_j) - H(S_i, S_j). \tag{A 4}$$

We can define the triple shared probability as (the triple joint probability p_{ijk} is constructed as before by tracing out all indices except i, j and k from the full joint probability $p(s_1 \cdots s_L)$)

$$p(i:j:k) = \frac{p_{ij}(s_i, s_j) p_{ik}(s_i, s_k) p_{jk}(s_j, s_k)}{p_i(s_i) p_j(s_j) p_k(s_k) p_{ijk}(s_i, s_j, s_k)}, \tag{A 5}$$

which implies the well-known formula for the triplet information

$$\begin{aligned} I(i:j:k) &= -\sum_{s_i, s_j, s_k} p_{ijk}(s_i, s_j, s_k) \log p(i:j:k) \\ &= H(S_i) + H(S_j) + H(S_k) - H(S_i, S_j) - H(S_i, S_k) - H(S_j, S_k) + H(S_i S_j S_k). \end{aligned} \tag{A 6}$$

The shared probability across all L sites (see [9, p. 58]) is defined in an analogous manner[8]:

$$p(s_1 : s_2 : \cdots : s_L) = \frac{(\Pi_{i<j} p_{ij})(\Pi_{i<j<k<m} p_{ijkm}) \cdots}{(\Pi_i p_i)(\Pi_{i<j<k} p_{ijk}) \cdots p(s_1, s_2 \cdots s_L)}. \tag{A 7}$$

Averaging (A 7) over the joint distribution of all variables gives the information decomposition theorem (2.5). Because of the symmetrical nature of (A 7), it is not difficult to see that it can be rewritten as

$$p(s_1 s_2 \cdots s_L) = \frac{(\Pi_{i<j} p_{i:j})(\Pi_{i<j<k<m} p_{i:j:k:m}) \cdots}{(\Pi_i p_i)(\Pi_{i<j<k} p_{i:j:k}) \cdots p(s_1 : s_2 : \cdots : s_L)}, \tag{A 8}$$

which, when averaged over $p(s_1 \cdots s_L)$, yields the entropy decomposition theorem (2.4).

[8]For simplicity, we have assumed here that L is odd in order to show the symmetry between joint and shared probabilities more clearly, but this is not necessary in principle, and is not implied in the entropy and information decomposition theorems.

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Appendix B. Relationship between multivariate correlation and 'connected information'

Using the notation in §2, we can write the 'multi-information' contained in a set of L potentially correlated random variables as

$$I_L = \sum_{i=1}^{L} H(S_i) - H(S_1 \cdots S_L). \tag{B 1}$$

The multi-information is a measure of the total correlation present in the system [11,41], and therefore may be composed of correlations of arbitrary order. In order to characterize the contribution of correlations of order k towards the multi-information [15], Schneidman *et al.* introduced the concept of 'connected information' of order k

$$I_C^{(k)} = H(S_1 \cdots S_{k-1}) - H(S_1 \cdots S_k). \tag{B 2}$$

Using the entropy decomposition theorem (2.4), it is not difficult to show that [15]

$$I_L = \sum_{k=2}^{L} I_C^{(k)}. \tag{B 3}$$

The second- and third-order connected information $I_C^{(2)}$ and $I_C^{(3)}$ are then simply the second and third term in the decomposition (2.6)

$$I_C^{(2)} = \sum_{i<j} I(S_i : S_j) \tag{B 4}$$

and

$$I_C^{(3)} = \sum_{i<j<k} I(S_i : S_j : S_k) \tag{B 5}$$

and so forth, which are just the multivariate correlations introduced above to calculate approximations to the true information content of composite variables.

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